

Lesson 01-A- Centre of Gravity and Moment of inertia (review)

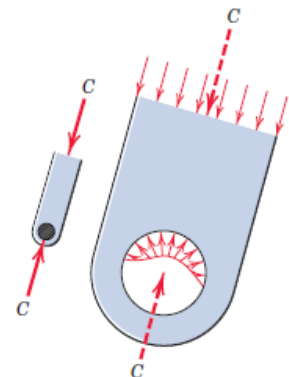
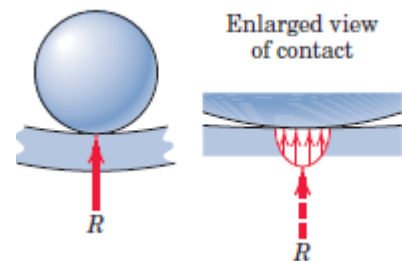
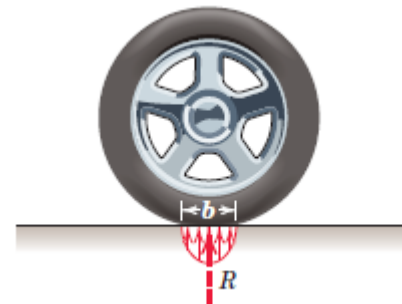
Distributed Forces Center of Area and Volume:

Distribution of Force:

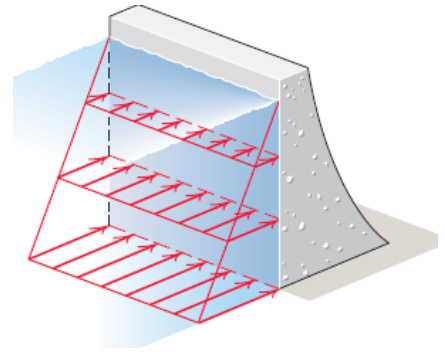
- In general we treated all forces as concentrated along their lines of action and at their points of application. This treatment provided a reasonable model for those forces.
- “CONCENTRATED” forces do not exist in the exact sense, since every external force applied mechanically to a body is distributed over a finite contact area, however small.

Examples:

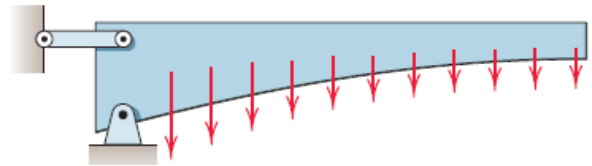
- The force exerted by the pavement on an automobile tire: if the dimension b of the contact area is negligible compared with the other pertinent dimensions, then we may replace the actual distributed contact forces by their resultant R
- Even the force of contact between a hardened steel ball and its race in a loaded ball bearing: is applied over a finite though extremely small contact area.
- The forces applied to a two-force member of a truss, are applied over an actual area of contact of the pin against the hole and internally across the cut section as shown.



- **Area Distribution:** When a force is distributed over an area, as with the hydraulic pressure of water against the inner face of a section of dam, the intensity is expressed as force per unit area. This intensity is called *pressure* for the action of fluid forces and *stress* for the internal distribution of forces in solids.
The basic unit for pressure or stress in SI is the newton per square meter (N/m^2),

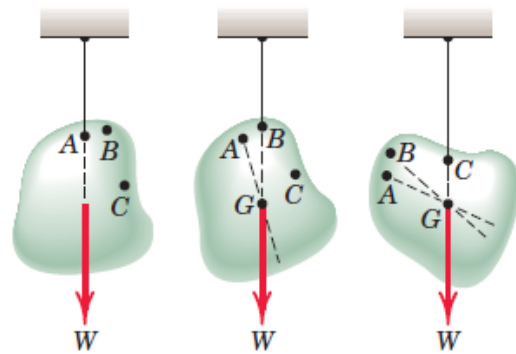


- **Volume Distribution:** A force which is distributed over the volume of a body is called a body force. The most common body force is the force of gravitational attraction, which acts on all elements of mass in a body.



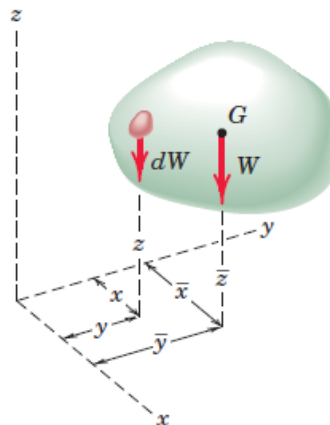
Center of Mass:

Consider a three-dimensional body of any size and shape, having a mass m . If we suspend the body, from any point such as A, B or C, the body will be in equilibrium under the action of the tension in the cord and the resultant W of the gravitational forces acting on all particles of the body



Determining the Center of Gravity:

To determine mathematically the location of the center of gravity of any body, we apply the principle of moments of gravitational forces. The moment of the resultant gravitational force W about any axis equals the sum of the moments about the same axis of the gravitational forces dW acting on all particles treated as infinitesimal elements of the body.



$$W \cdot \bar{x}' = \sum dw \cdot x \quad \dots\dots\dots (1)$$

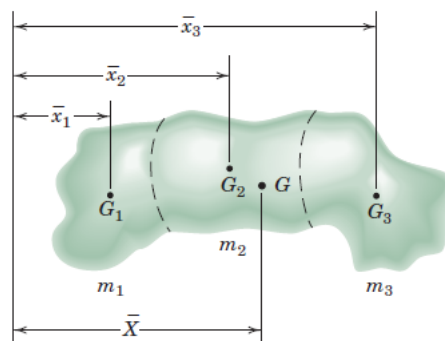
$$W \cdot \bar{y}' = \sum dw \cdot y \quad \dots\dots\dots (2)$$

$$W \cdot \bar{z}' = \sum dw \cdot z \quad \dots\dots\dots (3)$$

To find center of gravity or centroid of composite/ multiple bodies:

$$(m_1 + m_2 + m_3)\bar{X} = m_1\bar{x}_1 + m_2\bar{x}_2 + m_3\bar{x}_3$$

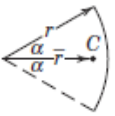
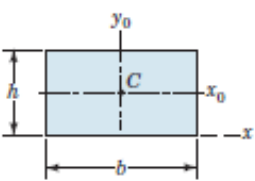
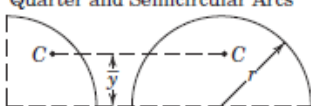
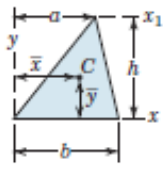
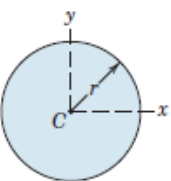
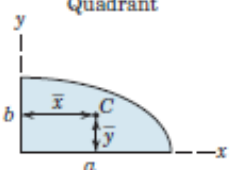
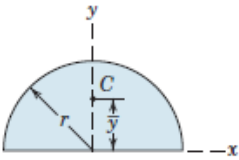
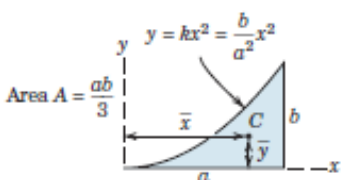
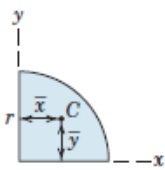
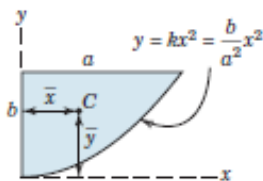
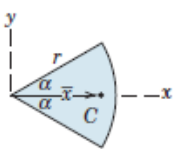
If the equation above is divided by the density, then $\bar{X}$ will represent the center of volume.



For a body of any number of parts and express the sums in condensed form to obtain the mass-center coordinates

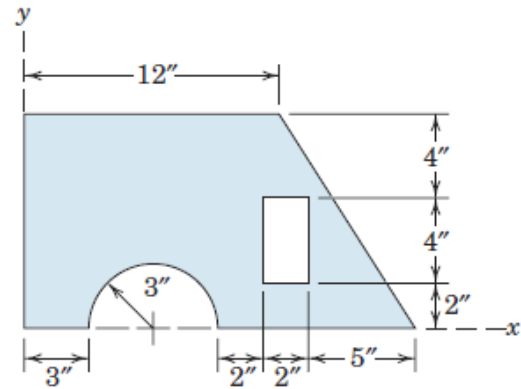
$$\bar{X} = \frac{\sum m\bar{x}}{\sum m} \quad \bar{Y} = \frac{\sum m\bar{y}}{\sum m} \quad \bar{Z} = \frac{\sum m\bar{z}}{\sum m}$$

Common Shapes and Curves Formulas for Center of Gravity:

FIGURE	CENTROID	FIGURE	CENTROID
Arc Segment 	$\bar{r} = \frac{r \sin \alpha}{\alpha}$	Rectangular Area 	—
Quarter and Semicircular Arcs 	$\bar{y} = \frac{2r}{\pi}$	Triangular Area 	$\bar{x} = \frac{a+b}{3}$ $\bar{y} = \frac{h}{3}$
Circular Area 	—	Area of Elliptical Quadrant 	$\bar{x} = \frac{4a}{3\pi}$ $\bar{y} = \frac{4b}{3\pi}$
Semicircular Area 	$\bar{y} = \frac{4r}{3\pi}$	Subparabolic Area 	$\bar{x} = \frac{3a}{4}$ $\bar{y} = \frac{3b}{10}$
Quarter-Circular Area 	$\bar{x} = \bar{y} = \frac{4r}{3\pi}$	Parabolic Area 	$\bar{x} = \frac{3a}{8}$ $\bar{y} = \frac{3b}{5}$
Area of Circular Sector 	$\bar{x} = \frac{2}{3} \frac{r \sin \alpha}{\alpha}$		

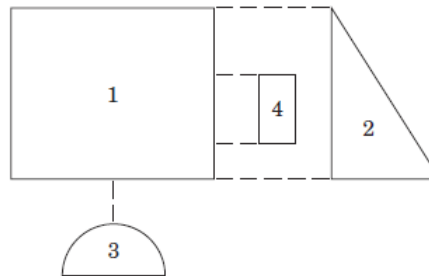
Example:

Locate the centroid of the shaded area.

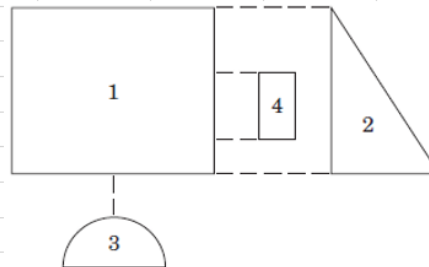
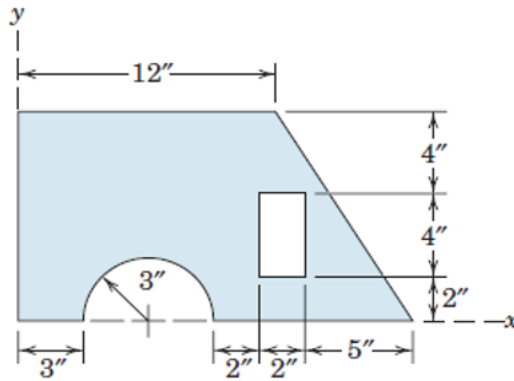


Solution:

using excel spread sheet:



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	A	B	C	D	E	F	G	H	I	J	K	L
1												
2	Part #	Area	x'	y'	x'.A	y'.A						
3												
4												
5												
6												
7												
8												
9												
10												
11												
12	Total											
13	(sum)											
14												
15												
16	$\bar{x} = \frac{\sum Ax_c}{\sum A}$			X' =								
17												
18	$\bar{y} = \frac{\sum Ay_c}{\sum A}$			Y' =								
19												
20												
21												
22												



PART	A in. ²	$\bar{x}$ in.	$\bar{y}$ in.	$\bar{x}A$ in. ³	$\bar{y}A$ in. ³
1	120	6	5	720	600
2	30	14	10/3	420	100
3	-14.14	6	1.273	-84.8	-18
4	-8	12	4	-96	-32
TOTALS	127.9			959	650

$$\left[\bar{X} = \frac{\Sigma A \bar{x}}{\Sigma A} \right]$$

$$\bar{X} = \frac{959}{127.9} = 7.50 \text{ in.}$$

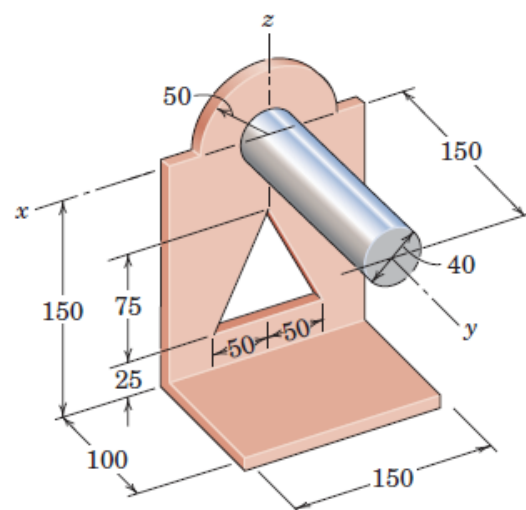
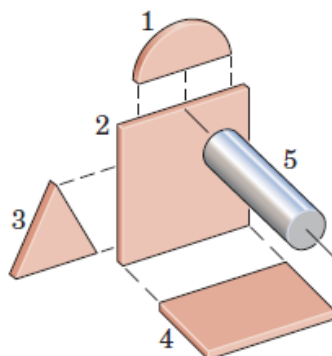
$$\left[\bar{Y} = \frac{\Sigma A \bar{y}}{\Sigma A} \right]$$

$$\bar{Y} = \frac{650}{127.9} = 5.08 \text{ in.}$$

Example:

Locate the center of mass of the bracket-and-shaft combination. The vertical face is made from sheet metal which has a mass of 25 kg/m². The material of the horizontal base has a mass of 40 kg/m², and the steel shaft has a density of 7.83 Mg/m³

Solution:



Dimensions in millimeters

Part #	Mass/ area or volume	Area or Volume (meter)	mass (kg)	x' (m)	y' (m)	z' (m)	x'.m (m.kg)	y'.m (m.kg)	z'.m (m.kg)
1.00	25.00	0.0039	0.098	0.000	0.000	0.021	0.0000	0.0000	0.0021
2.00	25.00	0.0225	0.563	0.000	0.000	-0.075	0.0000	0.0000	-0.0422
3.00	25.00	-0.0038	-0.094	0.000	0.000	-0.100	0.0000	0.0000	0.0094
4.00	40.00	0.0150	0.600	0.000	0.050	-0.150	0.0000	0.0300	-0.0900
5.00	7830.00	0.0002	1.475	0.000	0.075	0.000	0.0000	0.1106	0.0000
Total mass =			2.6420	sum=			0.0000	0.1406	-0.1207

$$\bar{X} = \frac{\sum x'.m}{\sum m}$$

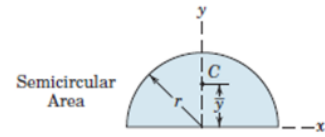
=	0.0000	mm
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$$\bar{Y} = \frac{\sum y'.m}{\sum m}$$

=	53.2307	mm
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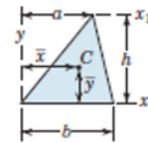
$$\bar{Z} = \frac{\sum z'.m}{\sum m}$$

=	-45.6953	mm
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$$\bar{y} = \frac{4r}{3\pi}$$

Triangular Area

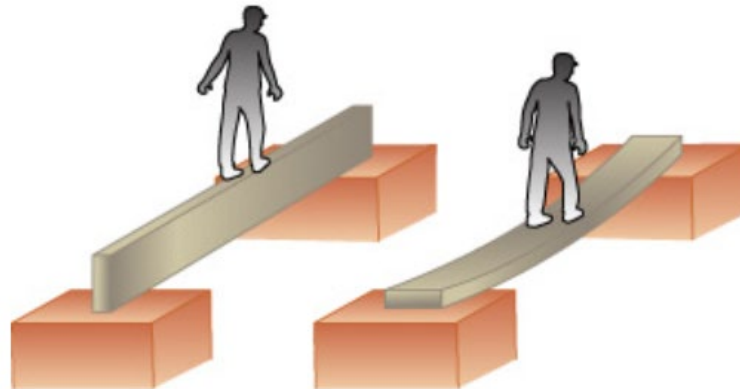


$$\bar{x} = \frac{a+b}{3}$$

$$\bar{y} = \frac{h}{3}$$

Area and Mass Moments of Inertia:

- The *second moment of area* is also known as the *moment of inertia* of a shape. It is directly related to the area of material in the cross-section and the displacement of that area from the centroid.
- Area moments of inertia are a measure of the distribution of a two-dimensional area around a particular axis. Fundamentally, the portions of a shape which are located farther from the axis have a greater affect than the parts which are closer.
- The primary application is in structural engineering and machine design where they are used to determine a structural member's stiffness.



Moment of Inertia:

The area moment of inertia defined by two equations and has units of $[\text{length}]^4$.

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

Polar Moment of Inertia:

The polar moment of inertia is defined as

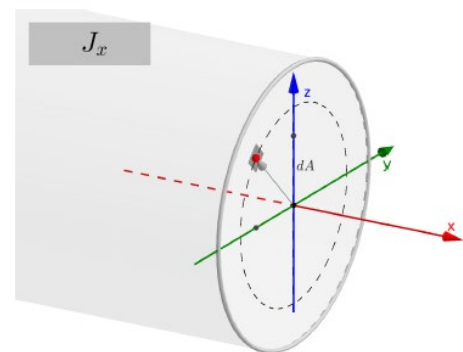
$$I_z = J_O = \int_A r^2 dA$$

Where:

$$I_z = I_x + I_y$$

The polar moment of inertia is another measure of the distribution of an area but, in this case, about a point at the origin rather than about an axis.

One important application of this value is to quantify the resistance of a shaft to torsion or twisting due to the shape of its cross-section.



Radius of Gyration:

Consider area A , which has rectangular moments of inertia I_x and I_y and a polar moment of inertia I_z about 0.

We now visualize this area as concentrated into a long narrow strip of area A a distance k_x from the x-axis,

By definition the moment of inertia of the strip about the x-axis will be the same as that of the original area if

$$k_x^2 A = I_x.$$

The distance k_x is called the radius of gyration of the area about the x-axis. Same concept applies on the other axis, hence, in summary:

$$\begin{array}{l} I_x = k_x^2 A \\ I_y = k_y^2 A \\ I_z = k_z^2 A \end{array} \quad \text{or} \quad \begin{array}{l} k_x = \sqrt{I_x/A} \\ k_y = \sqrt{I_y/A} \\ k_z = \sqrt{I_z/A} \end{array}$$

And:

$$k_z^2 = k_x^2 + k_y^2$$

Transfer of Axes:

The moment of inertia of an area about a non-centroidal axis may be easily expressed in terms of the moment of inertia about a parallel centroidal axis:

By definition, the moment of inertia of the element dA about the x-axis is:

$$dI_x = (y_0 + d_x)^2 dA$$

$$I_x = \int y_0^2 dA + 2d_x \int y_0 dA + d_x^2 \int dA$$

Hence:

$$\begin{array}{l} I_x = \bar{I}_x + Ad_x^2 \\ I_y = \bar{I}_y + Ad_y^2 \end{array}$$

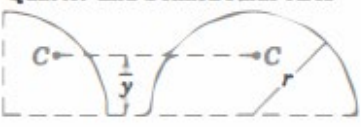
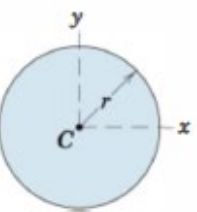
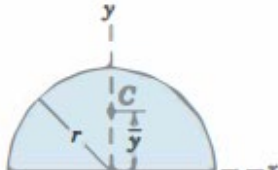
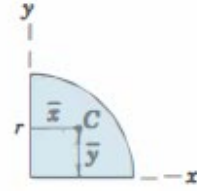
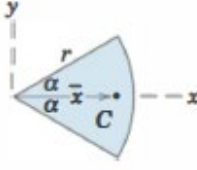
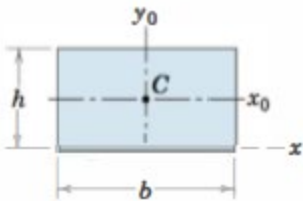
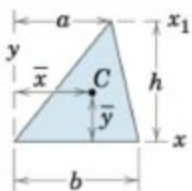
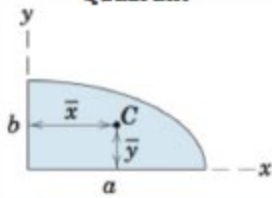
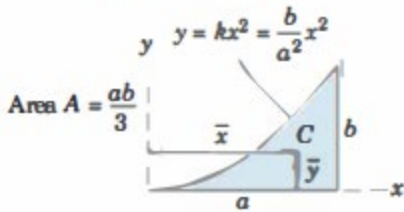
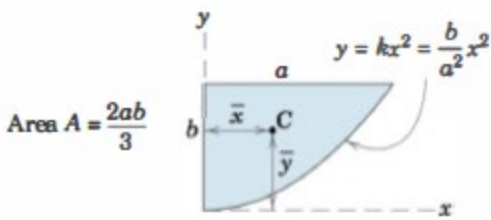
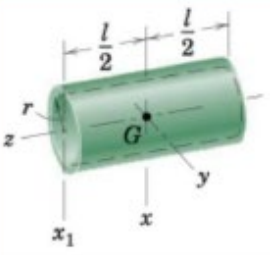
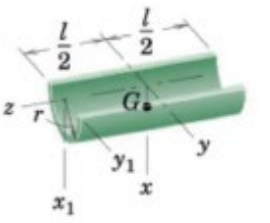
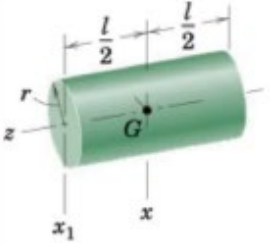
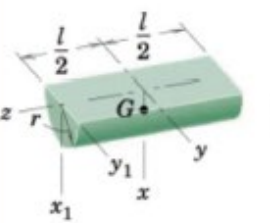
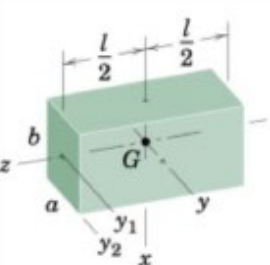
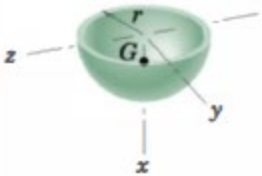
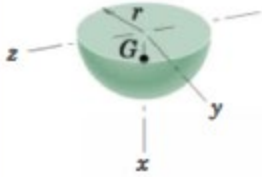
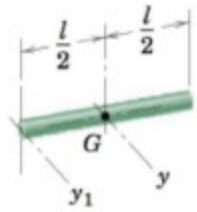
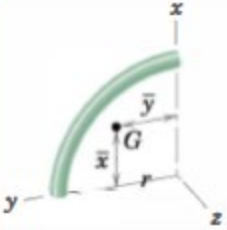
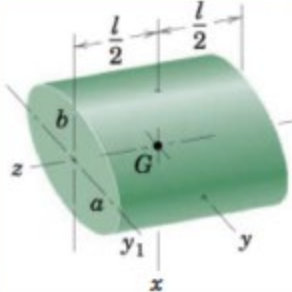
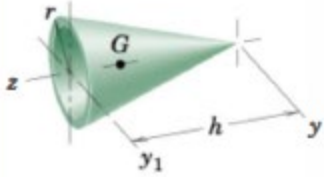
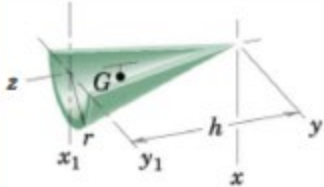
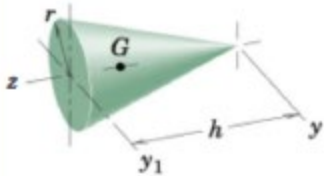
FIGURE	CENTROID	AREA MOMENTS OF INERTIA
Arc Segment 	$\bar{r} = \frac{r \sin \alpha}{\alpha}$	—
Quarter and Semicircular Arcs 	$\bar{y} = \frac{2r}{\pi}$	—
Circular Area 	—	$I_x = I_y = \frac{\pi r^4}{4}$ $I_z = \frac{\pi r^4}{2}$
Semicircular Area 	$\bar{y} = \frac{4r}{3\pi}$	$I_x = I_y = \frac{\pi r^4}{8}$ $\bar{I}_x = \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) r^4$ $I_z = \frac{\pi r^4}{4}$
Quarter-Circular Area 	$\bar{x} = \bar{y} = \frac{4r}{3\pi}$	$I_x = I_y = \frac{\pi r^4}{16}$ $\bar{I}_x = \bar{I}_y = \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) r^4$ $I_z = \frac{\pi r^4}{8}$
Area of Circular Sector 	$\bar{x} = \frac{2}{3} \frac{r \sin \alpha}{\alpha}$	$I_x = \frac{r^4}{4} \left(\alpha - \frac{1}{2} \sin 2\alpha \right)$ $I_y = \frac{r^4}{4} \left(\alpha + \frac{1}{2} \sin 2\alpha \right)$ $I_z = \frac{1}{2} r^4 \alpha$

FIGURE	CENTROID	AREA MOMENTS OF INERTIA
Rectangular Area 	—	$I_x = \frac{bh^3}{3}$ $\bar{I}_x = \frac{bh^3}{12}$ $\bar{I}_z = \frac{bh}{12}(b^2 + h^2)$
Triangular Area 	$\bar{x} = \frac{a+b}{3}$ $\bar{y} = \frac{h}{3}$	$I_x = \frac{bh^3}{12}$ $\bar{I}_x = \frac{bh^3}{36}$ $I_{x_1} = \frac{bh^3}{4}$
Area of Elliptical Quadrant 	$\bar{x} = \frac{4a}{3\pi}$ $\bar{y} = \frac{4b}{3\pi}$	$I_x = \frac{\pi ab^3}{16}, \quad \bar{I}_x = \left(\frac{\pi}{16} - \frac{4}{9\pi}\right)ab^3$ $I_y = \frac{\pi a^3 b}{16}, \quad \bar{I}_y = \left(\frac{\pi}{16} - \frac{4}{9\pi}\right)a^3 b$ $I_z = \frac{\pi ab}{16}(a^2 + b^2)$
Subparabolic Area 	$\bar{x} = \frac{3a}{4}$ $\bar{y} = \frac{3b}{10}$	$I_x = \frac{ab^3}{21}$ $I_y = \frac{a^3 b}{5}$ $I_z = ab\left(\frac{a^3}{5} + \frac{b^2}{21}\right)$
Parabolic Area 	$\bar{x} = \frac{3a}{8}$ $\bar{y} = \frac{3b}{5}$	$I_x = \frac{2ab^3}{7}$ $I_y = \frac{2a^3 b}{15}$ $I_z = 2ab\left(\frac{a^2}{15} + \frac{b^2}{7}\right)$

BODY	MASS CENTER	MASS MOMENTS OF INERTIA
 Circular Cylindrical Shell	—	$I_{xx} = \frac{1}{2}mr^2 + \frac{1}{12}ml^2$ $I_{x_1x_1} = \frac{1}{2}mr^2 + \frac{1}{3}ml^2$ $I_{zz} = mr^2$
 Half Cylindrical Shell	$\bar{x} = \frac{2r}{\pi}$	$I_{xx} = I_{yy}$ $= \frac{1}{2}mr^2 + \frac{1}{12}ml^2$ $I_{x_1x_1} = I_{y_1y_1}$ $= \frac{1}{2}mr^2 + \frac{1}{3}ml^2$ $I_{zz} = mr^2$ $\bar{I}_{zz} = \left(1 - \frac{4}{\pi^2}\right)mr^2$
 Circular Cylinder	—	$I_{xx} = \frac{1}{4}mr^2 + \frac{1}{12}ml^2$ $I_{x_1x_1} = \frac{1}{4}mr^2 + \frac{1}{3}ml^2$ $I_{zz} = \frac{1}{2}mr^2$
 Semicylinder	$\bar{x} = \frac{4r}{3\pi}$	$I_{xx} = I_{yy}$ $= \frac{1}{4}mr^2 + \frac{1}{12}ml^2$ $I_{x_1x_1} = I_{y_1y_1}$ $= \frac{1}{4}mr^2 + \frac{1}{3}ml^2$ $I_{zz} = \frac{1}{2}mr^2$ $\bar{I}_{zz} = \left(\frac{1}{2} - \frac{16}{9\pi^2}\right)mr^2$
 Rectangular Parallelepiped	—	$I_{xx} = \frac{1}{12}m(a^2 + l^2)$ $I_{yy} = \frac{1}{12}m(b^2 + l^2)$ $I_{zz} = \frac{1}{12}m(a^2 + b^2)$ $I_{y_1y_1} = \frac{1}{12}mb^2 + \frac{1}{3}ml^2$ $I_{y_2y_2} = \frac{1}{3}m(b^2 + l^2)$

BODY	MASS CENTER	MASS MOMENTS OF INERTIA
 Spherical Shell	—	$I_{zz} = \frac{2}{3}mr^2$
 Hemispherical Shell	$\bar{x} = \frac{r}{2}$	$I_{xx} = I_{yy} = I_{zz} = \frac{2}{3}mr^2$ $\bar{I}_{yy} = \bar{I}_{zz} = \frac{5}{12}mr^2$
 Sphere	—	$I_{zz} = \frac{2}{5}mr^2$
 Hemisphere	$\bar{x} = \frac{3r}{8}$	$I_{xx} = I_{yy} = I_{zz} = \frac{2}{5}mr^2$ $\bar{I}_{yy} = \bar{I}_{zz} = \frac{83}{320}mr^2$
 Uniform Slender Rod	—	$I_{yy} = \frac{1}{12}ml^2$ $I_{y_1y_1} = \frac{1}{3}ml^2$

BODY	MASS CENTER	MASS MOMENTS OF INERTIA
 Quarter-Circular Rod	$\bar{x} = \bar{y} = \frac{2r}{\pi}$	$I_{xx} = I_{yy} = \frac{1}{2}mr^2$ $I_{zz} = mr^2$
 Elliptical Cylinder	—	$I_{xx} = \frac{1}{4}ma^2 + \frac{1}{12}ml^2$ $I_{yy} = \frac{1}{4}mb^2 + \frac{1}{12}ml^2$ $I_{zz} = \frac{1}{4}m(a^2 + b^2)$ $I_{yy_1} = \frac{1}{4}mb^2 + \frac{1}{3}ml^2$
 Conical Shell	$\bar{z} = \frac{2h}{3}$	$I_{yy} = \frac{1}{4}mr^2 + \frac{1}{2}mh^2$ $I_{yy_1} = \frac{1}{4}mr^2 + \frac{1}{6}mh^2$ $I_{zz} = \frac{1}{2}mr^2$ $\bar{I}_{yy} = \frac{1}{4}mr^2 + \frac{1}{18}mh^2$
 Half Conical Shell	$\bar{x} = \frac{4r}{3\pi}$ $\bar{z} = \frac{2h}{3}$	$I_{xx} = I_{yy} = \frac{1}{4}mr^2 + \frac{1}{2}mh^2$ $I_{x_1x_1} = I_{y_1y_1} = \frac{1}{4}mr^2 + \frac{1}{6}mh^2$ $I_{zz} = \frac{1}{2}mr^2$ $\bar{I}_{zz} = \left(\frac{1}{2} - \frac{16}{9\pi^2}\right)mr^2$
 Right Circular Cone	$\bar{z} = \frac{3h}{4}$	$I_{yy} = \frac{3}{20}mr^2 + \frac{3}{8}mh^2$ $I_{y_1y_1} = \frac{3}{20}mr^2 + \frac{1}{10}mh^2$ $I_{zz} = \frac{3}{10}mr^2$ $\bar{I}_{yy} = \frac{3}{20}mr^2 + \frac{3}{80}mh^2$

PART	A mm ²	d _x mm	d _y mm	Ad _x ² mm ³	Ad _y ² mm ³	I _x mm ⁴	I _y mm ⁴
1	80(60)	30	40	4.32(10 ⁶)	7.68(10 ⁶)	$\frac{1}{12}(80)(60)^3$	$\frac{1}{12}(60)(80)^3$
2	$-\frac{1}{4}\pi(30)^2$	(60 – 12.73)	12.73	-1.579(10 ⁶)	-0.1146(10 ⁶)	$-\left(\frac{\pi}{16} - \frac{4}{9\pi}\right)30^4$	$-\left(\frac{\pi}{16} - \frac{4}{9\pi}\right)30^4$
3	$-\frac{1}{2}(40)(30)$	$\frac{30}{3}$	$\left(80 - \frac{40}{3}\right)$	-0.06(10 ⁶)	-2.67(10 ⁶)	$-\frac{1}{36}40(30)^3$	$-\frac{1}{36}(30)(40)^3$
TOTALS	3490			2.68(10 ⁶)	4.90(10 ⁶)	1.366(10 ⁶)	2.46(10 ⁶)
[I _x = Σ I _x + Σ Ad _x ²]	I _x = 1.366(10 ⁶) + 2.68(10 ⁶) = 4.05(10 ⁶) mm ⁴			Ansl.			
[I _y = Σ I _y + Σ Ad _y ²]	I _y = 2.46(10 ⁶) + 4.90(10 ⁶) = 7.36(10 ⁶) mm ⁴			Ansl.			